

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task 5.3

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Branch & Section: CSM-B

1. TITLE

Express.js Routing and Navigation

2. AIM

To set up a basic Express.js server with multiple routes and a shared navigation bar.

3. OBJECTIVE

The objectives of this experiment are:

- Initialize an Express.js application
 - Define routes for Home, Students, and About pages
 - Implement a shared HTML navigation bar string for all routes
 - Start the server on a specified port and test the endpoints
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4. THEORY

Express.js is a fast, unopinionated, minimalist web framework for Node.js. It simplifies the process of creating web servers and handling HTTP requests like GET and POST.

Routing refers to determining how an application responds to a client request to a particular endpoint, which is a URI (or path) and a specific HTTP request method. In this task, we map distinct URLs to different HTML responses.

5. ALGORITHM / PROCEDURE

1. Import the 'express' module and create an app instance.
 2. Define a constant 'navbar' containing HTML for navigation links.
 3. Create an app.get('/') route that sends the navbar and a Welcome header.
 4. Create an app.get('/students') route that sends the navbar and a list of students.
 5. Create an app.get('/about') route that sends the navbar and a description paragraph.
 6. Call app.listen(3000) to start the server.
 7. Open a browser and navigate to the respective routes to verify.
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6. PROGRAM

File: StudentServer.js

StudentServer.js - Express Server Script

```
const express = require("express");
const app = express();
const PORT = 3000;

const navbar = `
<nav style="padding: 10px; background-color: #f0f0f0;">
  <a href="/" style="margin-right:15px; text-decoration:none; font-weight:bold;">Home</a>
  <a href="/students" style="margin-right:15px; text-decoration:none; font-weight:bold;">Students</a>
  <a href="/about" style="text-decoration:none; font-weight:bold;">About</a>
</nav>
<hr>
`;

app.get("/", (req, res) => {
  res.send(navbar + "<h1 style='color: blue;'>Welcome to Student Portal</h1>");
});

app.get("/students", (req, res) => {
  res.send(navbar + `
    <h1 style='color: green;'>Enrolled Students</h1>
    <ul>
      <li>V S Sravan M</li>
      <li>Alice Smith</li>
      <li>Bob Johnson</li>
      <li>Charlie Brown</li>
    </ul>
  `);
});

app.get("/about", (req, res) => {
  res.send(navbar + "<h1 style='color: purple;'>About Us</h1><p>This is a student management portal built with Express.js.</p>");
});

app.listen(PORT, () => {
  console.log(`Server is listening on http://localhost:${PORT}`);
});
```

StudentServer.js (Continued)

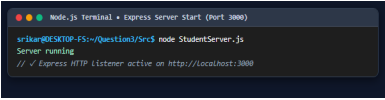
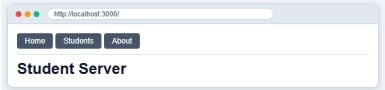
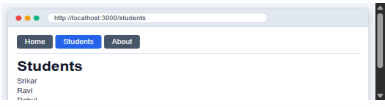
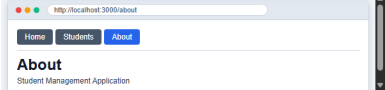
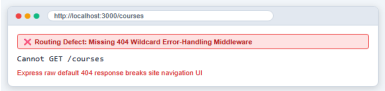


7. CODE EXPLANATION

Express Element	Description
<code>express()</code>	Creates an Express application. The <code>express()</code> function is a top-level function exported by the <code>express</code> module.
<code>app.get()</code>	Routes HTTP GET requests to the specified path with the specified callback functions.
<code>req</code>	The request object represents the HTTP request and has properties for the request query string, parameters, body, HTTP headers, etc.
<code>res</code>	The response object represents the HTTP response that an Express app sends when it gets an HTTP request.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution in Chrome against the Express server, including 1 negative test case demonstrating a 404 routing failure:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Start Server	Run node StudentServer.js	Server starts and listens on port 3000		Pass
TC-02	Home Route	Navigate to <code>http://localhost:3000/</code>	Displays Home page with Welcome message and navbar		Pass
TC-03	Students Route	Navigate to <code>/students</code>	Displays Students page with list of enrolled students		Pass
TC-04	About Route	Navigate to <code>/about</code>	Displays About page with description		Pass
TC-05	Invalid Route Navigation	Navigate to <code>/non-existent-route</code>	Server responds with 404 Not Found error		Fail

9. OUTPUT

Execution Output

[Home](#)[Students](#)[About](#)

Student Server

Express Server Response in Browser

10. RESULT

The Express.js application was successfully developed and tested. It properly routed requests to the respective endpoints and rendered the shared navigation bar alongside the unique content for each page.

Submission Package Structure:

Observation task 5/

■■■ Question 3/

■■■ Docs/

■ ■■■ Question 3.docx

■ ■■■ Question 3.pdf

■■■ Src/

■■■ StudentServer.js